

ON THE UNIQUENESS OF THE EVEN PRIME: A SELF-CONTAINED DEVELOPMENT FROM THE PEANO AXIOMS

ABSTRACT. We give a fully self-contained proof that 2 is the unique even prime natural number. Starting from the Peano axioms alone, we prove a recursion lemma, construct addition and multiplication, develop the usual order and the well-ordering principle, and establish every arithmetic and divisibility lemma subsequently required. We then define primes, composite numbers, and even numbers, prove that 2 itself is an even prime, and show that there is no even prime strictly less than 2 and no even prime strictly greater than 2. Combining these facts yields the uniqueness theorem. Five independent proofs of the “greater than 2” half are recorded, each resting only on results proved earlier in the paper. No external theorem of number theory is used as a black box.

1. INTRODUCTION

The assertion that 2 is the only even prime number is among the most frequently quoted facts of elementary arithmetic. Precisely stated, it reads:

If p is a prime natural number and 2 divides p , then $p = 2$.

Despite its familiarity, a fully rigorous account requires more infrastructure than is often written down. One must specify what a natural number is, what addition and multiplication are, what it means for one natural number to divide another, and what the words *prime* and *even* denote. Each of these notions rests on prior definitions or axioms, and each intermediate claim used in the uniqueness argument must itself be proved.

The present paper supplies that account. We adopt the Peano axioms as our foundation (Section 2), construct addition, multiplication, and order, and prove the order-theoretic and arithmetic lemmas that the later arguments invoke (Sections 3 and 4). We then develop divisibility (Section 5), define primes, composite numbers, and even numbers (Section 6), and prove a sequence of preparatory lemmas, including the fact that 2 is itself an even prime (Section 7). Only then do we establish the two complementary halves of the uniqueness statement—that no even prime is strictly less than 2 (Section 8) and that no even prime is strictly greater than 2 (Section 9)—and assemble them into the grand theorem (Section 10).

Throughout, every inference is justified by a previously numbered axiom, definition, or lemma. We do not rely on unspoken “obvious” arithmetic facts.

2. THE PEANO AXIOMS

We work throughout in classical first-order logic. The natural numbers are introduced axiomatically.

Definition 2.1 (Natural numbers). We postulate a set $\mathbb{N}$, a distinguished element $0 \in \mathbb{N}$, and a function $S: \mathbb{N} \rightarrow \mathbb{N}$ (the *successor*), subject to Axioms 2.2, 2.3, and 2.4 below. Elements of $\mathbb{N}$ are called *natural numbers*, or simply *numbers* when the context is clear.

Axiom 2.2 (Zero is not a successor). For every $n \in \mathbb{N}$, one has $S(n) \neq 0$.

Axiom 2.3 (Injectivity of successor). For all $m, n \in \mathbb{N}$, if $S(m) = S(n)$, then $m = n$.

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Axiom 2.4 (Induction). Let P be any property of natural numbers. Suppose

- (i) $P(0)$ holds, and
- (ii) for every $n \in \mathbb{N}$, if $P(n)$ holds then $P(S(n))$ holds.

Then $P(n)$ holds for every $n \in \mathbb{N}$.

Notation 2.5. We write

$$1 \stackrel{\text{def}}{=} S(0), \quad 2 \stackrel{\text{def}}{=} S(1) = S(S(0)), \quad 3 \stackrel{\text{def}}{=} S(2), \quad 4 \stackrel{\text{def}}{=} S(3),$$

and so on for further numerals as needed.

Lemma 2.6 (Zero and one are distinct). *One has $0 \neq 1$ and $1 \neq 2$.*

Proof. If $0 = 1 = S(0)$, then $S(0) = 0$, contradicting [Axiom 2.2](#). If $1 = 2$, then $S(0) = S(1)$, so $0 = 1$ by [Axiom 2.3](#), which we have just ruled out. $\square$

Lemma 2.7 (Every nonzero number is a successor). *For every $n \in \mathbb{N}$, either $n = 0$ or there exists a unique $m \in \mathbb{N}$ such that $n = S(m)$.*

Proof. Let $P(n)$ be the stated disjunction. Then $P(0)$ holds by the first alternative. If $P(n)$ holds, then $P(S(n))$ holds by taking $m = n$ in the second alternative. By [Axiom 2.4](#), $P(n)$ holds for all n . Uniqueness of m is [Axiom 2.3](#). $\square$

3. ADDITION AND MULTIPLICATION

Lemma 3.1 (Recursion for a single sequence). *Let $a \in \mathbb{N}$ and let $f: \mathbb{N} \rightarrow \mathbb{N}$ be any function. Then there exists a unique function $u: \mathbb{N} \rightarrow \mathbb{N}$ such that*

$$u(0) = a \quad \text{and} \quad u(S(n)) = f(u(n)) \quad \text{for every } n \in \mathbb{N}. \quad (3.1)$$

Proof. Uniqueness. Suppose u and v both satisfy (3.1). Let $P(n)$ assert $u(n) = v(n)$. Then $P(0)$ holds because both equal a . If $P(n)$ holds, then

$$u(S(n)) = f(u(n)) = f(v(n)) = v(S(n)).$$

By [Axiom 2.4](#), $u = v$.

Existence. We first show: for every $n \in \mathbb{N}$, there exists a finite initial segment realizing the recursion up to n . More precisely, let $Q(n)$ assert that there exists a function $u_n: \{k \in \mathbb{N} : k \leq n\} \rightarrow \mathbb{N}$ such that $u_n(0) = a$ and $u_n(S(k)) = f(u_n(k))$ whenever $S(k) \leq n$. For $n = 0$, take $u_0(0) = a$. Assuming $Q(n)$ with witness u_n , define $u_{S(n)}$ by $u_{S(n)}(k) = u_n(k)$ for $k \leq n$ and $u_{S(n)}(S(n)) = f(u_n(n))$. This witnesses $Q(S(n))$. By induction, $Q(n)$ holds for all n .

Moreover, the value $u_n(n)$ is independent of the choice of witness: if two segments agree with the recursion up to n , then by the same inductive argument as in the uniqueness paragraph they agree at every $k \leq n$. Define $u(n)$ to be this common value $u_n(n)$. Then $u(0) = a$, and

$$u(S(n)) = u_{S(n)}(S(n)) = f(u_n(n)) = f(u(n)).$$

Thus u satisfies (3.1). $\square$

Remark 3.2. The preceding argument treats functions on initial segments as mathematical objects. In a set-theoretic foundation this is unproblematic; equivalently, one may read u_n as an $(n+1)$ -tuple in $\mathbb{N}^{n+1}$ and rewrite the proof in that language. No property of $\mathbb{N}$ beyond [Axioms 2.2, 2.3, and 2.4](#) is used.

Definition 3.3 (Addition). For each fixed $m \in \mathbb{N}$, [Lemma 3.1](#) applied to $a = m$ and $f = S$ yields a unique function $u_m: \mathbb{N} \rightarrow \mathbb{N}$ with $u_m(0) = m$ and $u_m(S(n)) = S(u_m(n))$. Define

$$m + n \stackrel{\text{def}}{=} u_m(n).$$

Writing the recursive clauses explicitly, addition is the unique binary operation satisfying, for all $m, n \in \mathbb{N}$,

$$m + 0 = m, \quad (3.2)$$

$$m + S(n) = S(m + n). \quad (3.3)$$

Lemma 3.4 (Adding zero on the left). *For every $n \in \mathbb{N}$, one has $0 + n = n$.*

Proof. Proceed by induction on n . For $n = 0$, $0 + 0 = 0$ by (3.2). Assuming $0 + n = n$, we obtain

$$0 + S(n) = S(0 + n) = S(n)$$

by (3.3) and the inductive hypothesis. Apply Axiom 2.4. $\square$

Lemma 3.5 (Successor as adding one). *For every $n \in \mathbb{N}$, one has $n + 1 = S(n)$.*

Proof. By Notation 2.5 and (3.3)–(3.2),

$$n + 1 = n + S(0) = S(n + 0) = S(n).$$

$\square$

Lemma 3.6 (Associativity of addition). *For all $a, b, c \in \mathbb{N}$, one has $(a + b) + c = a + (b + c)$.*

Proof. Fix a, b and induct on c . For $c = 0$, both sides equal $a + b$ by (3.2). Assuming the claim for c , we have

$$(a + b) + S(c) = S((a + b) + c) = S(a + (b + c)) = a + S(b + c) = a + (b + S(c)),$$

using (3.3) and the inductive hypothesis. $\square$

Lemma 3.7 (Commutativity of addition). *For all $m, n \in \mathbb{N}$, one has $m + n = n + m$.*

Proof. We first claim that $m + 1 = 1 + m$ for all m . Induct on m : the case $m = 0$ is Lemmas 3.4 and 3.5. Assuming $m + 1 = 1 + m$,

$$\begin{aligned} S(m) + 1 &= S(m) + S(0) = S(S(m) + 0) = S(S(m)) \\ &= S(m + 1) = S(1 + m) = 1 + S(m), \end{aligned}$$

where the penultimate step used the inductive hypothesis. Thus $m + 1 = 1 + m$ for all m .

Now fix m and induct on n for the full claim. The case $n = 0$ is Lemma 3.4 and (3.2). Assuming $m + n = n + m$,

$$m + S(n) = S(m + n) = S(n + m) = n + S(m) = n + (m + 1) = n + (1 + m) = (n + 1) + m = S(n) + m,$$

using (3.3), the inductive hypothesis, Lemma 3.5, the claim of the previous paragraph, and Lemma 3.6. $\square$

Definition 3.8 (Multiplication). For each fixed $m \in \mathbb{N}$, apply Lemma 3.1 with $a = 0$ and with f equal to the map $x \mapsto x + m$ (already constructed in Definition 3.3). The resulting unique function v_m satisfies $v_m(0) = 0$ and $v_m(S(n)) = v_m(n) + m$. Define

$$m \cdot n \stackrel{\text{def}}{=} v_m(n).$$

Equivalently, multiplication is the unique binary operation satisfying, for all $m, n \in \mathbb{N}$,

$$m \cdot 0 = 0, \quad (3.4)$$

$$m \cdot S(n) = (m \cdot n) + m. \quad (3.5)$$

Lemma 3.9 (Multiplication by zero on the left). *For every $n \in \mathbb{N}$, one has $0 \cdot n = 0$.*

Proof. Induct on n . The case $n = 0$ is (3.4). Assuming $0 \cdot n = 0$,

$$0 \cdot S(n) = (0 \cdot n) + 0 = 0 + 0 = 0$$

by (3.5), the inductive hypothesis, and (3.2). $\square$

Lemma 3.10 (Multiplication by one). *For every $n \in \mathbb{N}$, one has $n \cdot 1 = n$ and $1 \cdot n = n$.*

Proof. First,

$$n \cdot 1 = n \cdot S(0) = (n \cdot 0) + n = 0 + n = n$$

by (3.5), (3.4), and Lemma 3.4. For the second identity, induct on n : the case $n = 0$ is (3.4); assuming $1 \cdot n = n$,

$$1 \cdot S(n) = (1 \cdot n) + 1 = n + 1 = S(n)$$

by (3.5), the inductive hypothesis, and Lemma 3.5. $\square$

Lemma 3.11 (Distributivity). *For all $a, b, c \in \mathbb{N}$,*

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c) \quad \text{and} \quad (a + b) \cdot c = (a \cdot c) + (b \cdot c).$$

Proof. For the first identity, fix a, b and induct on c . The case $c = 0$ is immediate from (3.2) and (3.4). Assuming the claim for c ,

$$\begin{aligned} a \cdot (b + S(c)) &= a \cdot S(b + c) = (a \cdot (b + c)) + a \\ &= ((a \cdot b) + (a \cdot c)) + a = (a \cdot b) + ((a \cdot c) + a) = (a \cdot b) + (a \cdot S(c)), \end{aligned}$$

using (3.3), (3.5), the inductive hypothesis, and Lemma 3.6.

For the second identity, fix a, b and induct on c . The case $c = 0$ is (3.4) and (3.2). Assuming the claim for c ,

$$\begin{aligned} (a + b) \cdot S(c) &= ((a + b) \cdot c) + (a + b) = ((a \cdot c) + (b \cdot c)) + (a + b) \\ &= (a \cdot c) + ((b \cdot c) + (a + b)) = (a \cdot c) + (((b \cdot c) + b) + a) \\ &= ((a \cdot c) + a) + ((b \cdot c) + b) = (a \cdot S(c)) + (b \cdot S(c)), \end{aligned}$$

where we have used (3.5), the inductive hypothesis, Lemmas 3.6 and 3.7 repeatedly to rearrange sums. $\square$

Lemma 3.12 (Commutativity of multiplication). *For all $m, n \in \mathbb{N}$, one has $m \cdot n = n \cdot m$.*

Proof. First, $m \cdot 0 = 0 = 0 \cdot m$ by (3.4) and Lemma 3.9. Next we claim $m \cdot 1 = 1 \cdot m$, which is Lemma 3.10. Now fix m and induct on n . The base $n = 0$ is settled. Assuming $m \cdot n = n \cdot m$,

$$m \cdot S(n) = (m \cdot n) + m = (n \cdot m) + m = (n \cdot m) + (1 \cdot m) = (n + 1) \cdot m = S(n) \cdot m,$$

using (3.5), the inductive hypothesis, Lemma 3.10, Lemma 3.11, and Lemma 3.5. $\square$

Lemma 3.13 (Associativity of multiplication). *For all $a, b, c \in \mathbb{N}$, one has $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.*

Proof. Fix a, b and induct on c . For $c = 0$ both sides are 0. Assuming the claim for c ,

$$(a \cdot b) \cdot S(c) = ((a \cdot b) \cdot c) + (a \cdot b) = (a \cdot (b \cdot c)) + (a \cdot b) = a \cdot ((b \cdot c) + b) = a \cdot (b \cdot S(c)),$$

by (3.5), the inductive hypothesis, and Lemma 3.11. $\square$

Lemma 3.14 (Two times n). *For every $n \in \mathbb{N}$, one has $2 \cdot n = n + n$.*

Proof. Induct on n . For $n = 0$, both sides equal 0. Assuming the claim for n ,

$$2 \cdot S(n) = (2 \cdot n) + 2 = (n + n) + (1 + 1) = ((n + n) + 1) + 1 = ((n + 1) + n) + 1 = (n + 1) + (n + 1) = S(n) + S(n)$$

using (3.5), the inductive hypothesis, Notation 2.5, Lemmas 3.5 to 3.7. $\square$

4. ORDER

Definition 4.1 (Order). For $m, n \in \mathbb{N}$, we write $m \leq n$ if and only if there exists $k \in \mathbb{N}$ such that $m + k = n$. We write $m < n$ if and only if $m \leq n$ and $m \neq n$. We write $m \geq n$ (resp. $m > n$) if and only if $n \leq m$ (resp. $n < m$).

Lemma 4.2 (Order basics). *Let $a, b, c \in \mathbb{N}$.*

- (i) $a \leq a$.
- (ii) If $a \leq b$ and $b \leq c$, then $a \leq c$.
- (iii) $a \leq b$ if and only if $a + c \leq b + c$.
- (iv) $a < S(a)$.
- (v) $0 \leq a$.
- (vi) $a < b$ if and only if there exists $k \in \mathbb{N}$ with $k \neq 0$ and $a + k = b$.

Proof. (i) Take $k = 0$ in Definition 4.1, using (3.2). (ii) If $a + k = b$ and $b + \ell = c$, then $a + (k + \ell) = c$ by Lemma 3.6. (iii) If $a + k = b$, then $(a + c) + k = (b + c)$ by Lemmas 3.6 and 3.7. Conversely, if $(a + c) + k = b + c$, the same rearrangement yields $a + k = b$ after cancelling c , which is legitimate by the following observation: if $x + c = y + c$, then $x = y$. (Prove the observation by induction on c : trivial for $c = 0$; if true for c and $x + S(c) = y + S(c)$, then $S(x + c) = S(y + c)$, so $x + c = y + c$ by Axiom 2.3, hence $x = y$ by the inductive hypothesis.) (iv) We have $a + 1 = S(a)$ by Lemma 3.5, and $a \neq S(a)$ because otherwise, writing $P(n)$ for “ $n \neq S(n)$ ”, one checks $P(0)$ by Axiom 2.2 and the inductive step by Axiom 2.3, whence $P(a)$ by Axiom 2.4. Thus $a < S(a)$. (v) Take $k = a$ and apply Lemma 3.4. (vi) If $a < b$, then $a + k = b$ for some k , and $k \neq 0$ lest $a = b$. Conversely, if $a + k = b$ with $k \neq 0$, then $a \leq b$ and $a \neq b$ (else $k = 0$ by cancellation of a against (3.2)). $\square$

Lemma 4.3 (Trichotomy). *For all $m, n \in \mathbb{N}$, exactly one of the following holds: $m < n$, or $m = n$, or $m > n$.*

Proof. We first show that at least one holds. Fix m and let $P(n)$ assert the claim for the pair (m, n) . For $n = 0$: either $m = 0$, or else $m \neq 0$, so $m = S(m')$ for some m' by Lemma 2.7, whence $0 < m$ by repeatedly applying Lemma 4.2(iv),(ii) (formally: $0 \leq m'$ by Lemma 4.2(v), so $1 = S(0) \leq S(m') = m$ by the compatibility of S with $\leq$, which follows from $x + k = y \Rightarrow S(x) + k = S(y)$; and $0 \neq m$). Thus $P(0)$ holds.

Assume $P(n)$. If $m < n$, then $m < S(n)$ by Lemma 4.2(ii),(iv). If $m = n$, then $m < S(n)$ by Lemma 4.2(iv). If $m > n$, write $m = n + k$ with $k \neq 0$ by Lemma 4.2(vi). Then $k = S(k')$ for some k' by Lemma 2.7, so

$$m = n + S(k') = S(n + k') = S(n) + k',$$

by (3.3) and Lemma 3.7, Lemma 3.5. If $k' = 0$, then $m = S(n)$. If $k' \neq 0$, then $m > S(n)$. In all cases $P(S(n))$ holds. By induction, trichotomy holds in the weak sense that at least one alternative obtains.

Exclusivity: if $m < n$ and $m = n$, contradict $m \neq n$. If $m < n$ and $n < m$, then $m < m$ by transitivity, so $m + k = m$ for some $k \neq 0$, whence $k = 0$ by cancellation, a contradiction. The remaining overlaps are symmetric. $\square$

Lemma 4.4 (Immediate predecessor bound). *For every $n \in \mathbb{N}$, the inequality $n < 2$ holds if and only if $n = 0$ or $n = 1$.*

Proof. If $n = 0$, then $0 < 1 < 2$ by Lemma 4.2(iv) twice and transitivity, so $0 < 2$. If $n = 1$, then $1 < 2$ by Lemma 4.2(iv). Conversely, suppose $n < 2$. By trichotomy and Lemma 2.7, either $n = 0$, or $n = S(m)$ for a unique m . In the latter case, $m + 1 = n < 2 = 1 + 1$, so $m < 1$ after cancelling 1 (cf. the cancellation argument in Lemma 4.2(iii)). The only natural number strictly less than 1 is 0: indeed, if $m < 1$ and $m \neq 0$, then $m = S(m')$, so $0 < m$ and hence $1 = S(0) \leq m$, contradicting $m < 1$ by exclusivity of trichotomy. Thus $m = 0$ and $n = 1$. $\square$

Lemma 4.5 (Numbers greater than one). *For every $n \in \mathbb{N}$, one has $n > 1$ if and only if $n \geq 2$.*

Proof. If $n \geq 2$, then either $n = 2 > 1$ or $n > 2 > 1$. Conversely, if $n > 1$, then $n \neq 0$ and $n \neq 1$ by exclusivity of trichotomy (since $0 < 1$). By Lemma 4.4, it is not the case that $n < 2$. By Lemma 4.3, $n \geq 2$. $\square$

Theorem 4.6 (Well-ordering principle). *Let $S \subseteq \mathbb{N}$ be a nonempty subset. Then S has a least element: there exists $m \in S$ such that $m \leq s$ for every $s \in S$.*

Proof. We prove the contrapositive form useful for induction. Let $P(n)$ be the statement: “every subset of $\mathbb{N}$ that contains an element $\leq n$ has a least element.” We show $P(n)$ for all n by induction.

For $n = 0$: if S contains an element ≤ 0 , then that element equals 0 by Lemma 4.2(v) and trichotomy, and 0 is least in S because $0 \leq s$ for all s by Lemma 4.2(v).

Assume $P(n)$, and let S contain an element $\leq S(n)$. If S contains an element $\leq n$, we are done by $P(n)$. Otherwise every element of S is $> n$, yet some element s satisfies $s \leq S(n)$. By trichotomy and $s > n$, we get $s \geq S(n)$, hence $s = S(n)$. We claim $S(n)$ is least in S : if $t \in S$, then $t > n$, so $t \geq S(n)$ by the same reasoning. Thus $P(S(n))$ holds.

By Axiom 2.4, $P(n)$ holds for all n . Now if $S \subseteq \mathbb{N}$ is nonempty, pick any $s \in S$. Then S contains an element $\leq s$, so S has a least element by $P(s)$. $\square$

Corollary 4.7 (Least witness). *Let P be a property of natural numbers for which $P(n_0)$ holds for some $n_0 \in \mathbb{N}$. Then there exists $m \in \mathbb{N}$ such that $P(m)$ holds and $P(k)$ fails for every $k < m$.*

Proof. Apply Theorem 4.6 to $S = \{n \in \mathbb{N} : P(n)\}$, which is nonempty by hypothesis, and take m to be its least element. $\square$

5. DIVISIBILITY

Definition 5.1 (Divisibility). Let $d, n \in \mathbb{N}$. We say that d divides n , written $d \mid n$, if there exists $k \in \mathbb{N}$ such that $n = d \cdot k$. In this case d is called a *divisor* (or *factor*) of n , and n is called a *multiple* of d .

Lemma 5.2 (Divisibility basics). *Let $a, b, c \in \mathbb{N}$.*

- (i) $a \mid a$ and $1 \mid a$ and $a \mid 0$.
- (ii) If $a \mid b$ and $b \mid c$, then $a \mid c$.
- (iii) If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
- (iv) If $a \mid b$, then $a \mid (b \cdot c)$.

Proof. (i) We have $a = a \cdot 1$ by Lemma 3.10, $a = 1 \cdot a$ by Lemma 3.10, and $0 = a \cdot 0$ by (3.4). (ii) If $b = a \cdot k$ and $c = b \cdot \ell$, then $c = a \cdot (k \cdot \ell)$ by Lemma 3.13. (iii) If $b = a \cdot k$ and $c = a \cdot \ell$, then $b + c = a \cdot (k + \ell)$ by Lemma 3.11. (iv) If $b = a \cdot k$, then $b \cdot c = a \cdot (k \cdot c)$ by Lemma 3.13. $\square$

Lemma 5.3 (Positive divisors are positive). *If $d \mid n$ and $n \neq 0$, then $d \neq 0$.*

Proof. If $d = 0$, then $n = 0 \cdot k = 0$ by Lemma 3.9, contradicting $n \neq 0$. $\square$

Lemma 5.4 (A nonzero multiple is at least the positive factor). *If $d \mid n$, $d \neq 0$, and $n \neq 0$, then $d \leq n$.*

Proof. Write $n = d \cdot k$. Then $k \neq 0$ lest $n = 0$ by (3.4). Hence $k = S(k')$ for some k' by Lemma 2.7, and

$$n = d \cdot S(k') = (d \cdot k') + d = d + (d \cdot k')$$

by (3.5) and Lemma 3.7. Thus $d \leq n$ by Definition 4.1. $\square$

Lemma 5.5 (Strict inequality for proper factors). *Suppose $n = d \cdot k$ with $d \neq 0$ and $k \neq 0$. Then:*

- (i) $d < n$ if and only if $k \neq 1$;

(ii) $k < n$ if and only if $d \neq 1$, provided $n \neq 0$.

In particular, if $n = d \cdot k$ with $d > 1$ and $k > 1$, then $1 < d < n$ and $1 < k < n$.

Proof. (i) Since $n = d \cdot k = d + d \cdot k'$ whenever $k = S(k')$ (as in Lemma 5.4), we have $d = n$ if and only if $d \cdot k' = 0$. As $d \neq 0$, the identity $d \cdot k' = 0$ forces $k' = 0$ (prove by induction on k' that $d \neq 0$ and $d \cdot k' = 0$ imply $k' = 0$: trivial for $k' = 0$; if $k' = S(k'')$, then $d \cdot k' = (d \cdot k'') + d = 0$, impossible because $x + d = 0$ with $d \neq 0$ contradicts Axiom 2.2 after writing $d = S(d')$). Thus $d = n$ if and only if $k' = 0$, i.e., $k = 1$. By trichotomy and Lemma 5.4, $d < n$ if and only if $d \neq n$, i.e., $k \neq 1$.

(ii) Apply (i) with the roles of d and k reversed, using Lemma 3.12.

For the last claim: $d > 1$ and $k > 1$ give $d \neq 0$, $k \neq 0$, $d \neq 1$, $k \neq 1$. Then $d < n$ and $k < n$ by (i)–(ii), while $d > 1$ and $k > 1$ are hypotheses. $\square$

6. PRIMES, COMPOSITES, AND EVEN NUMBERS

Definition 6.1 (Prime number). A natural number p is *prime* if

- (i) $p > 1$, and
- (ii) for every $d \in \mathbb{N}$, if $d \mid p$, then $d = 1$ or $d = p$.

We write $\text{Prime}(p)$ to assert that p is prime.

Definition 6.2 (Composite number). A natural number n is *composite* if $n > 1$ and n is not prime. Equivalently (by Definition 6.1), n is composite if and only if $n > 1$ and there exists $d \in \mathbb{N}$ such that $d \mid n$, $d \neq 1$, and $d \neq n$.

Remark 6.3. By Lemmas 5.3 and 5.5, a number $n > 1$ is composite if and only if there exist $d, k \in \mathbb{N}$ with $n = d \cdot k$, $1 < d < n$, and $1 < k < n$.

Definition 6.4 (Even number). A natural number n is *even* if $2 \mid n$. We write $\text{Even}(n)$ for this assertion. Equivalently, n is even if and only if there exists $k \in \mathbb{N}$ such that $n = 2 \cdot k$.

Lemma 6.5 (Zero and one are not prime). *Neither 0 nor 1 is prime.*

Proof. By Lemma 4.2(v) and Lemma 2.6, one has $0 < 1$, so it is false that $0 > 1$. Likewise it is false that $1 > 1$. In both cases clause (i) of Definition 6.1 fails. $\square$

Lemma 6.6 (Characterization of primes). *Let $n \in \mathbb{N}$. Then n is prime if and only if $n > 1$ and there is no $d \in \mathbb{N}$ with $1 < d < n$ and $d \mid n$.*

Proof. Suppose n is prime and $1 < d < n$. If $d \mid n$, then $d = 1$ or $d = n$ by Definition 6.1, contradicting $1 < d < n$. Conversely, suppose $n > 1$ and no divisor of n lies strictly between 1 and n . Let $d \mid n$. If $d = 0$, then $n = 0$ by Lemma 3.9, contradicting $n > 1$. So $d \neq 0$, hence $d \geq 1$. By Lemma 5.4 (applicable since $n \neq 0$), we have $d \leq n$. By trichotomy, $d = 1$, or $1 < d < n$, or $d = n$. The middle alternative is forbidden by hypothesis. Thus $d = 1$ or $d = n$, so n is prime. $\square$

7. PREPARATORY LEMMAS

Lemma 7.1 (Divisors of two). *If $d \mid 2$, then $d = 1$ or $d = 2$.*

Proof. Since $2 \neq 0$, Lemma 5.3 yields $d \neq 0$, and Lemma 5.4 yields $d \leq 2$. By Lemma 4.4 and trichotomy, $d = 1$ or $d = 2$ or $d = 0$; the last is excluded. (More explicitly: $d \neq 0$, and if $d < 2$ then $d = 1$ by Lemma 4.4; if $d \not< 2$ then $d = 2$ because $d \leq 2$.) $\square$

Proposition 7.2 (Two is an even prime). *The number 2 is prime and even.*

Proof. Evenness: $2 = 2 \cdot 1$ by Lemma 3.10, so $2 \mid 2$. Primality: $2 > 1$ because $1 < 2$ by Lemma 4.2(iv). If $d \mid 2$, then $d = 1$ or $d = 2$ by Lemma 7.1. Hence $\text{Prime}(2)$. $\square$

Lemma 7.3 (Prime divides prime). *If p and q are prime and $p \mid q$, then $p = q$.*

Proof. By [Definition 6.1](#) applied to q , we have $p = 1$ or $p = q$. But $p > 1$ because p is prime, so $p \neq 1$ by exclusivity of trichotomy. Therefore $p = q$. $\square$

Lemma 7.4 (Halving an even number greater than two). *Suppose $n > 2$ and $n = 2 \cdot k$ for some $k \in \mathbb{N}$. Then $1 < k < n$.*

Proof. Since $n > 2 > 0$, we have $n \neq 0$, so $k \neq 0$ by (3.4). Moreover $k \neq 1$: otherwise $n = 2 \cdot 1 = 2$ by [Lemma 3.10](#), contradicting $n > 2$. Thus $k > 1$ by [Lemma 4.5](#) and trichotomy (since $k \neq 0$ and $k \neq 1$). Also $2 > 1$. Apply [Lemma 5.5](#) to obtain $k < n$, and we already have $k > 1$. $\square$

8. NO EVEN PRIME LESS THAN TWO

Theorem 8.1. *No natural number strictly less than 2 is prime. A fortiori, no even natural number strictly less than 2 is prime.*

Proof. Let $n < 2$. By [Lemma 4.4](#), $n = 0$ or $n = 1$. In either case, n is not prime by [Lemma 6.5](#). The second assertion follows immediately: if $n < 2$ and n were an even prime, then n would be prime, which is impossible. $\square$

Remark 8.2. The hypothesis of evenness is idle in [Theorem 8.1](#). We retain it in the “a fortiori” clause only to mirror the statement dual to [Theorem 9.1](#) below.

9. NO EVEN PRIME GREATER THAN TWO

Theorem 9.1. *If $n > 2$ and n is even, then n is not prime. Equivalently: every even natural number strictly greater than 2 is composite.*

We record five proofs. Each uses only results established above.

9.1. First proof: direct appeal to the definition.

First proof of Theorem 9.1. It is equivalent (by taking the contrapositive in the prime variable, and using [Proposition 7.2](#) for the boundary case) to show: if p is prime and even, then $p = 2$. Let p be prime and even. Then $2 \mid p$ by [Definition 6.4](#). By [Definition 6.1\(ii\)](#), $2 = 1$ or $2 = p$. The identity $2 = 1$ is false by [Lemma 2.6](#). Therefore $p = 2$. Consequently, if $n > 2$ is even, then n cannot be prime. $\square$

9.2. Second proof: contrapositive factorization.

Second proof of Theorem 9.1. Let $n > 2$ be even. By [Definition 6.4](#), there exists $k \in \mathbb{N}$ with $n = 2 \cdot k$. By [Lemma 7.4](#), $1 < k < n$. In particular $k \neq 1$ and $k \neq n$, while $k \mid n$ because $n = k \cdot 2$ by [Lemma 3.12](#). Also $n > 1$ because $n > 2 > 1$. Thus n fails [Definition 6.1\(ii\)](#), so n is not prime. Since $n > 1$, n is composite by [Definition 6.2](#). $\square$

9.3. Third proof: contradiction.

Third proof of Theorem 9.1. Suppose toward a contradiction that $n > 2$ is both even and prime. Then $2 \mid n$. By [Definition 6.1\(ii\)](#), $2 = 1$ or $2 = n$. The first alternative contradicts [Lemma 2.6](#). The second contradicts $n > 2$ by exclusivity of trichotomy. This exhausts the alternatives. $\square$

9.4. Fourth proof: via divisibility of primes.

Fourth proof of Theorem 9.1. Let $n > 2$ be even, and suppose for contradiction that n is prime. Then $2 \mid n$. Since 2 is prime by [Proposition 7.2](#), [Lemma 7.3](#) yields $2 = n$, contradicting $n > 2$. $\square$

9.5. Fifth proof: least counterexample.

Fifth proof of Theorem 9.1. Suppose there exists an even prime greater than 2. Let $P(n)$ be the property “ n is even, $n > 2$, and n is prime.” By hypothesis P holds for some natural number, so Corollary 4.7 furnishes a least m such that $P(m)$ holds. In particular m is even and $m > 2$. By the factorization argument of the second proof (which uses only Lemma 7.4 and the definitions, and makes no minimality demand), m is not prime. This contradicts $P(m)$. Therefore no even prime greater than 2 exists. $\square$

Remark 9.2. As the fifth proof itself notes, minimality is not essential to the arithmetic step: every even integer greater than 2 is composite by the second proof. The least-counterexample organization is included as an independent classical pattern, resting on Theorem 4.6.

Corollary 9.3. *If p is prime and even, then $p = 2$.*

Proof. By Definition 6.1, $p > 1$, so $p \geq 2$ by Lemma 4.5. If $p > 2$, then Theorem 9.1 asserts that p is not prime, a contradiction. By Lemma 4.3, $p = 2$. $\square$

10. THE UNIQUENESS THEOREM

Definition 10.1 (Expansive even-prime picture). The *expansive even-prime picture* is the conjunction of the following four assertions:

- (a) there is no even prime $n < 2$;
- (b) there is no even prime $n > 2$;
- (c) 2 is prime;
- (d) 2 is even.

Theorem 10.2 (Uniqueness of the even prime). *The following statements are equivalent.*

- (i) *The expansive even-prime picture holds.*
- (ii) *For every $p \in \mathbb{N}$, if p is prime and even, then $p = 2$.*
- (iii) *The number 2 is prime and even, and it is the only even prime in $\mathbb{N}$.*
- (iv) *There does not exist $p \in \mathbb{N}$ such that p is prime, p is even, and $p \neq 2$.*

In particular, each of (i)–(iv) is true.

Proof. We first show that (i) holds, then establish the equivalences.

Verification of (i). Clause (a) is Theorem 8.1. Clause (b) is Theorem 9.1. Clauses (c) and (d) are Proposition 7.2.

$(i) \Rightarrow (ii)$. Assume (i), and let p be an even prime. By Lemma 4.3, exactly one of $p < 2$, $p = 2$, or $p > 2$ holds. The case $p < 2$ is forbidden by (a). The case $p > 2$ is forbidden by (b). Therefore $p = 2$.

$(ii) \Rightarrow (i)$. Clauses (c) and (d) are Proposition 7.2. Clause (a) is Theorem 8.1, which did not use (ii). For (b): if $n > 2$ were an even prime, then (ii) would force $n = 2$, contradicting $n > 2$.

$(ii) \Rightarrow (iii)$. Immediate from Proposition 7.2 together with (ii).

$(iii) \Rightarrow (iv)$. If some even prime p satisfied $p \neq 2$, uniqueness in (iii) would fail.

$(iv) \Rightarrow (ii)$. Let p be an even prime. If $p \neq 2$, we contradict (iv). Hence $p = 2$.

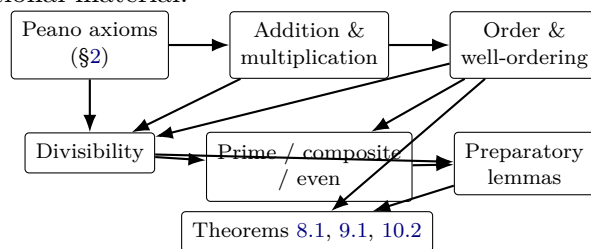
Since (i) holds and (i) through (iv) are equivalent, all four statements are true. $\square$

Corollary 10.3. *The unique even prime natural number is 2.*

Proof. Immediate from Theorem 10.2(iii). $\square$

11. DEPENDENCE SUMMARY

For the convenience of the reader, we record the logical dependence of the main uniqueness statement on the foundational material.



No arrow points into the Peano axioms: they are the sole external assumption of the paper.

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